

lab09(SSU)

October 1, 2025

1 Lab 09: Groups

Helpful Resource: - [Python Reference](#): Cheat sheet of helpful array & table methods!

Recommended Readings:

- [Visualizing Numerical Distributions](#)
- [Functions and Tables](#)

Please complete this notebook by filling in the cells provided. **Before you begin, execute the cell below to setup the notebook by importing some helpful libraries.** Each time you start your server, you will need to execute this cell again.

```
[4]: !apt-get install texlive texlive-xetex texlive-latex-extra pandoc
      !pip install pypandoc
      !pip install datascience
```

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

pandoc is already the newest version (2.9.2.1-3ubuntu2).

pandoc set to manually installed.

The following additional packages will be installed:

```
dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntaxtex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
teckit tex-common tex-gyre texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-recommended
texlive-pictures texlive-plain-generic tipa xfonts-encodings xfonts-utils
```

Suggested packages:

```
fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
```

```
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
default-jre-headless tipa-doc
```

The following NEW packages will be installed:

```
dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
teckit tex-common tex-gyre texlive texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-extra
texlive-latex-recommended texlive-pictures texlive-plain-generic
texlive-xetex tipa xfonts-encodings xfonts-utils
```

0 upgraded, 54 newly installed, 0 to remove and 38 not upgraded.

Need to get 5,783 kB/182 MB of archives.

After this operation, 571 MB of additional disk space will be used.

```
Ign:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9-common
all 9.55.0~dfsg1-0ubuntu5.12
```

```
Ign:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9 amd64
9.55.0~dfsg1-0ubuntu5.12
```

```
Err:1 http://security.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9-common
all 9.55.0~dfsg1-0ubuntu5.12
```

```
404 Not Found [IP: 91.189.91.81 80]
```

```
Err:2 http://security.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9 amd64
9.55.0~dfsg1-0ubuntu5.12
```

```
404 Not Found [IP: 91.189.91.81 80]
```

```
E: Failed to fetch http://security.ubuntu.com/ubuntu/pool/main/g/ghostscript/lib
gs9-common_9.55.0%7edfsg1-0ubuntu5.12_all.deb 404 Not Found [IP: 91.189.91.81
80]
```

```
E: Failed to fetch http://security.ubuntu.com/ubuntu/pool/main/g/ghostscript/lib
gs9_9.55.0%7edfsg1-0ubuntu5.12_amd64.deb 404 Not Found [IP: 91.189.91.81 80]
```

E: Unable to fetch some archives, maybe run apt-get update or try with --fix-missing?

Requirement already satisfied: py pandoc in /usr/local/lib/python3.12/dist-packages (1.15)

Requirement already satisfied: data science in /usr/local/lib/python3.12/dist-packages (0.18.0)

Requirement already satisfied: folium>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from data science) (0.20.0)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from data science) (75.2.0)

Requirement already satisfied: matplotlib>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from data science) (3.10.0)

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datascience) (2.2.2)

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from datascience) (1.16.2)

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from datascience) (2.0.2)

Requirement already satisfied: ipython in /usr/local/lib/python3.12/dist-packages (from datascience) (7.34.0)

Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (from datascience) (5.24.1)

Requirement already satisfied: branca in /usr/local/lib/python3.12/dist-packages (from datascience) (0.8.1)

Requirement already satisfied: jinja2>=2.9 in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (3.1.6)

Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (2.32.4)

Requirement already satisfied: xyzservices in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (2025.4.0)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (4.60.0)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (1.4.9)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (25.0)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (3.2.4)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (2.9.0.post0)

Requirement already satisfied: jedi>=0.16 in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.19.2)

Requirement already satisfied: decorator in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (4.4.2)

Requirement already satisfied: pickleshare in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.7.5)

Requirement already satisfied: traitlets>=4.2 in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (5.7.1)

Requirement already satisfied: prompt-toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (3.0.52)

Requirement already satisfied: pygments in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (2.19.2)

Requirement already satisfied: backcall in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.2.0)

Requirement already satisfied: matplotlib-inline in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.1.7)

Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (4.9.0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datascience) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datascience) (2025.2)

Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from plotly->datascience) (8.5.0)

Requirement already satisfied: parso<0.9.0,>=0.8.4 in /usr/local/lib/python3.12/dist-packages (from jedi>=0.16->ipython->datascience) (0.8.5)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2>=2.9->folium>=0.9.1->datascience) (3.0.2)

Requirement already satisfied: ptyprocess>=0.5 in /usr/local/lib/python3.12/dist-packages (from pexpect>4.3->ipython->datascience) (0.7.0)

Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-packages (from prompt-toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0->ipython->datascience) (0.2.13)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.0.0->datascience) (1.17.0)

Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->folium>=0.9.1->datascience) (3.4.3)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->folium>=0.9.1->datascience) (3.10)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->folium>=0.9.1->datascience) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->folium>=0.9.1->datascience) (2025.8.3)

```
[5]: # Connect Google Drive to Colab so you can access your files
from google.colab import drive
drive.mount('/content/drive')

import os
```

```
os.chdir('/content/drive/MyDrive/Colab Notebooks/')
```

Mounted at /content/drive

```
[6]: # Run this cell to set up the notebook, but please don't change it.

# These lines import the Numpy and Datascience modules.
import numpy as np
from datascience import *

# These lines do some fancy plotting magic.
import matplotlib
%matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')

import warnings
warnings.simplefilter('ignore', FutureWarning)
warnings.filterwarnings("ignore")
```

1.1 1. Based on the Lecture Activity

```
[7]: drinks = Table(['Drink', 'Cafe', 'Price'])
drinks = drinks.with_rows([
    ['Milk Tea', 'Asha', 5.5],
    ['Espresso', 'Strada', 1.75],
    ['Latte', 'Strada', 3.25],
    ['Espresso', "FSM", 2]
])
drinks
```

```
[7]: Drink    | Cafe    | Price
Milk Tea | Asha    | 5.5
Espresso | Strada  | 1.75
Latte    | Strada  | 3.25
Espresso | FSM     | 2
```

```
[8]: discounts = Table().with_columns(
    'Coupon % off', make_array(10, 25, 5),
    'Location', make_array('Asha', 'Strada', 'Asha')
)
discounts
```

```
[8]: Coupon % off | Location
10      | Asha
25      | Strada
5       | Asha
```

Question 1 Join drinks and discounts based on Cafe/Location.

```
[11]: combined = drinks.join("Cafe", discounts, "Location")
      combined
```

```
[11]: Cafe | Drink | Price | Coupon % off
      Asha | Milk Tea | 5.5 | 10
      Asha | Milk Tea | 5.5 | 5
      Strada | Espresso | 1.75 | 25
      Strada | Latte | 3.25 | 25
```

Question 2 Calculate the lowest rate you can get for each drink using the available coupons.

```
[61]: discount_frac = combined.column("Coupon % off") / 100
      combined.with_column("apply coupon", combined.column("Price") * (1 -
      ↪ discount_frac)).group("Drink", min)
```

```
[61]: Drink | Cafe min | Price min | Coupon % off min | apply coupon min
      Espresso | Strada | 1.75 | 25 | 1.3125
      Latte | Strada | 3.25 | 25 | 2.4375
      Milk Tea | Asha | 5.5 | 5 | 4.95
```

Question 3 What happens if I run the following? How many rows will it produce? Why?

```
[62]: drinks.join('Cafe', drinks, 'Cafe')
```

```
[62]: Cafe | Drink | Price | Drink_2 | Price_2
      Asha | Milk Tea | 5.5 | Milk Tea | 5.5
      FSM | Espresso | 2 | Espresso | 2
      Strada | Espresso | 1.75 | Espresso | 1.75
      Strada | Espresso | 1.75 | Latte | 3.25
      Strada | Latte | 3.25 | Espresso | 1.75
      Strada | Latte | 3.25 | Latte | 3.25
```

The code `drinks.join('Cafe', drinks, 'Cafe')` performs a self-join of the drinks table. In a join operation, when the join key matches, every possible pair of rows from the left table and the right table is combined. Specifically, the original table has 'Strada' appearing twice, so joining on 'Cafe' will produce $2 \times 2 = 4$ rows for 'Strada'. Since 'Asha' and 'FSM' each appear once, they will generate $1 \times 1 = 1$ row each. Therefore, the resulting table will have a total of $4 + 1 + 1 = 6$ rows.

1.2 2. Bike Sharing

```
[26]: trip = Table.read_table('./DS/trip.csv')
      trip.show(3)
```

<IPython.core.display.HTML object>

```
[31]: a=trip.column(3)
      items=np.unique(a, return_counts=True)
      for k,v in zip(items[0],items[1]):
          print(k,v)
```

```
2nd at Folsom 5396
2nd at South Park 6065
2nd at Townsend 9418
5th at Howard 5324
Adobe on Almaden 392
Arena Green / SAP Center 430
Beale at Market 5621
Broadway St at Battery St 5288
California Ave Caltrain Station 288
Castro Street and El Camino Real 828
Civic Center BART (7th at Market) 5367
Clay at Battery 3587
Commercial at Montgomery 4060
Cowper at University 403
Davis at Jackson 3639
Embarcadero at Bryant 5248
Embarcadero at Folsom 5331
Embarcadero at Sansome 10048
Embarcadero at Vallejo 3530
Evelyn Park and Ride 611
Franklin at Maple 54
Golden Gate at Polk 2545
Grant Avenue at Columbus Avenue 5667
Harry Bridges Plaza (Ferry Building) 12201
Howard at 2nd 4754
Japantown 633
MLK Library 798
Market at 10th 8006
Market at 4th 6527
Market at Sansome 7632
Mechanics Plaza (Market at Battery) 4269
Mezes Park 173
Mountain View Caltrain Station 2626
Mountain View City Hall 1135
Palo Alto Caltrain Station 815
Park at Olive 263
Paseo de San Antonio 582
Post at Kearny 3122
Powell Street BART 6371
Powell at Post (Union Square) 4351
Redwood City Caltrain Station 732
Redwood City Medical Center 131
```

Redwood City Public Library 68
 Rengstorff Avenue / California Street 373
 Ryland Park 650
 SJSU - San Salvador at 9th 371
 SJSU 4th at San Carlos 335
 San Antonio Caltrain Station 689
 San Antonio Shopping Center 743
 San Francisco Caltrain (Townsend at 4th) 17496
 San Francisco Caltrain 2 (330 Townsend) 15230
 San Francisco City Hall 1466
 San Jose City Hall 439
 San Jose Civic Center 504
 San Jose Diridon Caltrain Station 3393
 San Mateo County Center 91
 San Pedro Square 951
 San Salvador at 1st 311
 Santa Clara County Civic Center 344
 Santa Clara at Almaden 1379
 South Van Ness at Market 3941
 Spear at Folsom 3832
 St James Park 560
 Stanford in Redwood City 386
 Steuart at Market 9743
 Temporary Transbay Terminal (Howard at Beale) 9992
 Townsend at 7th 9691
 University and Emerson 391
 Washington at Kearny 1852
 Yerba Buena Center of the Arts (3rd @ Howard) 3575

Question 1 Create pivot table for Start Station and End Station. What value is being shown in each cell?

```
[34]: pivotTable = trip.pivot('Start Station', 'End Station')
      pivotTable
      #The value shown in each cell of this table represents the count of bike
      ↪rentals for trips originating at the corresponding 'Start Station' and
      ↪ending at the corresponding 'End Station'.
      #In other words, it is the aggregated number of trips between a specific
      ↪starting point and a specific destination.
```

```
[34]: End Station          | 2nd at Folsom | 2nd at South Park | 2nd at
Townsend | 5th at Howard | Adobe on Almaden | Arena Green / SAP Center | Beale
at Market | Broadway St at Battery St | California Ave Caltrain Station | Castro
Street and El Camino Real | Civic Center BART (7th at Market) | Clay at Battery
| Commercial at Montgomery | Cowper at University | Davis at Jackson |
Embarcadero at Bryant | Embarcadero at Folsom | Embarcadero at Sansome |
Embarcadero at Vallejo | Evelyn Park and Ride | Franklin at Maple | Golden Gate
at Polk | Grant Avenue at Columbus Avenue | Harry Bridges Plaza (Ferry Building)
```


| Howard at 2nd | Japantown | MLK Library | Market at 10th | Market at 4th |
 Market at Sansome | Mechanics Plaza (Market at Battery) | Mezes Park | Mountain
 View Caltrain Station | Mountain View City Hall | Palo Alto Caltrain Station |
 Park at Olive | Paseo de San Antonio | Post at Kearny | Powell Street BART |
 Powell at Post (Union Square) | Redwood City Caltrain Station | Redwood City
 Medical Center | Redwood City Public Library | Rengstorff Avenue / California
 Street | Ryland Park | SJSU - San Salvador at 9th | SJSU 4th at San Carlos | San
 Antonio Caltrain Station | San Antonio Shopping Center | San Francisco Caltrain
 (Townsend at 4th) | San Francisco Caltrain 2 (330 Townsend) | San Francisco City
 Hall | San Jose City Hall | San Jose Civic Center | San Jose Diridon Caltrain
 Station | San Mateo County Center | San Pedro Square | San Salvador at 1st |
 Santa Clara County Civic Center | Santa Clara at Almaden | South Van Ness at
 Market | Spear at Folsom | St James Park | Stanford in Redwood City | Steuart at
 Market | Temporary Transbay Terminal (Howard at Beale) | Townsend at 7th |
 University and Emerson | Washington at Kearny | Yerba Buena Center of the Arts
 (3rd @ Howard)

2nd at Folsom		53		127		396
82	0	0		27		
8		0		0		
31		45		40		
0	8	24		27		
36	13	0		0		
4	17			372		
41	0	0	129	84		180
26		0	0			
0	0		0	0		0
39	47	46				0
0		0		0		
0	0	0		0		0
0		474				285
18	0	0	0			0
0	0	0	0		0	
0	32		37		0	
0	30		207			
297	0	6		20		
2nd at South Park		149	107		39	
140	0	0		136		
53	0			0		
43		53	111			
0	19	31		131		
67	72	0			0	
34	43		428			
337	0	0	46	103		978
70		0	0			
0	0		0		0	
194	47	76			0	
0		0		0		

0	0	0	0
0	367	226	
30	0	0	0
0	0	0	0
0	49	84	0
0	266	213	
92	0	52	109
2nd at Townsend	349	109	164
57	0	0	424
238	0		0
56	207	102	
0	258	276	441
315	215	0	0
16	213	1837	
185	0	0	83
129		0	176
0	0	0	569
85	71	97	0
0	0	0	0
0	0	0	0
0	613		207
16	0	0	0
0	0	0	0
0	32	323	0
0	1685	489	
324	0	49	97
5th at Howard	80	126	63
136	0	0	43
103	0		0
167	47	71	
0	29	47	31
125	41	0	0
64	48	288	
353	0	0	251
55		0	128
0	0	0	121
78	142	112	0
0	0	0	0
0	0	0	0
0	432		1244
42	0	0	0
0	0	0	0
0	65	56	0
0	145	558	
135	0	30	218
Adobe on Almaden	0	0	0
0	19	6	0

0			0				0
0				0		0	
0		0		0			0
0		0			0		0
0		0			0		
0	14		3		0	0	0
0				0		0	
0		0				0	24
0	0			0			0
0		0				0	
8	6			12			0
0			0				0
0		13		13			176
0		18		4		3	
4		0			0		11
0		0		0			
0	0			0		0	
Arena Green / SAP Center			0		0		0
0	9		80			0	
0		0				0	
0			0		0		
0		0		0		0	
0		0			1		0
0	0				0		
0	14	5		0		0	0
0			0		0		
0	0	0			0		14
0			0				0
0		0			0		
20	2			4		0	
0		0					0
0		4		17			7
0		32		9		21	
154		0			0		5
0		0		0			
0	0			0		0	
Beale at Market			71		50		116
48	0		0			69	
462		0				0	
159			60		63		
0		167		114		27	
401		109			0		0
43		286				58	
46	0	0		181		124	115
21			0		0		
0		0			0		0
33	161		133				0

0		0		0		0
0	0		0		0	
0		423				191
15		0	0			0
0		0	0		0	
0		168		60		0
0		12	112			
15	0		43		21	
Broadway St at Battery St		39	59		189	
99	0	0		745		
115		0		0		
49		227	188			
0	131	167		47		
201	30	0			0	
4	58		125			
32	0	0	25	79	246	
162		0	0			
0	0		0		0	
108	50	150			0	
0		0		0		
0	0	0		0		
0		438			295	
8	0	0		0		
0	0	0		0	0	
0	11	72			0	
0	233	504				
30	0	59		30		
California Ave Caltrain Station	0	0		0		
0	0	0		0		
0		68		1		
0		0	0			
24	0	0	0	0		
0	0		0		0	
0	0		0			
0	0	0	0	0	0	
0		0	2			
4	138	31		0		
0	0	0		0		
0	0		7			
0	0	0		11		
7	0			0		
0	0	0		0		
0	0	0		0		
0	0	0		0		
1	0	0				
0	48	0		0		
Castro Street and El Camino Real	0	0		0		

```

| 0          | 0          | 0          | 0
| 0          |          | 0          |          | 66
| 0          |          | 0          |          | 0
| 0          | 0          |          | 0          | 0
| 0          | 0          |          | 9          | 0
| 0          | 0          |          | 0          | 0
| 0          | 0          | 0          | 0          | 0
| 0          | 0          |          | 636        | 0
| 34         |          | 0          | 1          | 0
| 0          | 0          |          | 0          | 0
| 0          |          | 0          |          | 10
| 0          | 0          |          | 0          | 3
| 16         |          | 0          |          | 0
| 0          |          | 0          |          | 0
| 0          |          | 0          |          | 0
| 0          |          | 0          |          | 0
| 0          |          | 0          |          | 0
| 0          | 1          |          | 0          | 0
... (60 rows omitted)

```

Question 2 Create pivot table showing average duration of the trips between the Start Station and End Station ?

```

[36]: averagePivot = trip.pivot('Start Station', 'End Station', values='Duration',
    ↪collect=np.mean)
averagePivot

```

```

[36]: End Station          | 2nd at Folsom | 2nd at South Park | 2nd at
Townsend | 5th at Howard | Adobe on Almaden | Arena Green / SAP Center | Beale
at Market | Broadway St at Battery St | California Ave Caltrain Station | Castro
Street and El Camino Real | Civic Center BART (7th at Market) | Clay at Battery
| Commercial at Montgomery | Cowper at University | Davis at Jackson |
Embarcadero at Bryant | Embarcadero at Folsom | Embarcadero at Sansome |
Embarcadero at Vallejo | Evelyn Park and Ride | Franklin at Maple | Golden Gate
at Polk | Grant Avenue at Columbus Avenue | Harry Bridges Plaza (Ferry Building)
| Howard at 2nd | Japantown | MLK Library | Market at 10th | Market at 4th |
Market at Sansome | Mechanics Plaza (Market at Battery) | Mezes Park | Mountain
View Caltrain Station | Mountain View City Hall | Palo Alto Caltrain Station |
Park at Olive | Paseo de San Antonio | Post at Kearny | Powell Street BART |
Powell at Post (Union Square) | Redwood City Caltrain Station | Redwood City
Medical Center | Redwood City Public Library | Rengstorff Avenue / California
Street | Ryland Park | SJSU - San Salvador at 9th | SJSU 4th at San Carlos | San
Antonio Caltrain Station | San Antonio Shopping Center | San Francisco Caltrain
(Townsend at 4th) | San Francisco Caltrain 2 (330 Townsend) | San Francisco City
Hall | San Jose City Hall | San Jose Civic Center | San Jose Diridon Caltrain
Station | San Mateo County Center | San Pedro Square | San Salvador at 1st |
Santa Clara County Civic Center | Santa Clara at Almaden | South Van Ness at

```

Market | Spear at Folsom | St James Park | Stanford in Redwood City | Steuart at
 Market | Temporary Transbay Terminal (Howard at Beale) | Townsend at 7th |
 University and Emerson | Washington at Kearny | Yerba Buena Center of the Arts
 (3rd @ Howard)

2nd at Folsom		2678.47		405.961		337.677
463.354	0		0		432.815	
881.75		0			0	
744.774		493.844		482.75		
0	865.875		636.042		573.852	
2659.83	12791.4		0		0	
789	2473.59			505.489		
197.537	0	0	728.357	433.917		
323.517	353.769			0	0	
0	0			0	0	
442.051	611.043		485.891		0	
0		0		0		
0	0		0		0	
0		473.416				
484.032		1096.17			0	
0	0			0		
0	0	0			0	
919.562	355.865	0		0		
1195.13	311.667				641.572	
0	709		401.95			
2nd at South Park		264.678		1432.28		280.641
535.379	0	0		527.485		
685.679	0			0		
776.14		617.226		611.694		
0	553.158	329.097		385.29		
1081.19	643.722	0			0	
655.441	1096.02			876.544		
219.881	0	0	787.826	631.087		
301.815	379.857			0	0	
0	0			0	0	
292.423	915.745	607.118			0	
0		0		0		
0	0	0		0		
0	285.659					
540.549		883.6		0		
0	0			0		
0	0	0			0	
777.347	355.274	0		0		
559.526	322.615				466.054	
0	703.346		384.367			
2nd at Townsend		431.779		538.037		2510.52
601.789	0	0		741.637		
784.836	0			0		

764.018		787.333		852.676	
0	776.155	430.822		540.664	
1265.01	1847.95	0		0	
992.125	926.61		664.486		
560.784	0	0	979.964	1227.18	
516.301	944.14		0	0	
0	0		0	0	
1054.07	2778.37	733.557		0	
0		0		0	
0	0	0		0	
0	354.341				
317.546		1165.62		0	
0	0		0		
0	0	0			0
1115.56	430.18	0		0	
559.707	535.278			567.623	
0	1146.53	603.99			
5th at Howard		499.175	681.151	3167.84	
6305.24	0	0		932.884	
891.777	0			0	
466.036		705.979	649.38		
0	1020	636.532		606.194	
1315.94	3204.76	0		0	
481.438	832.542		727.91		
425.048	0	0	735.884	343.273	
410.421	634.945		0	0	
0	0		0	0	
551.744	913.725	410.955		0	
0		0		0	
0	0	0		0	
0	428.13				
401.162		686.905		0	
0	0		0		
0	0	0			0
550.785	593.036	0		0	
836.117	405.131			576.815	
0	1024.3	241.362			
Adobe on Almaden	0	0		0	
0	2865.63	3475.33		0	
0	0			0	
0		0	0		
0	0	0		0	
0	0	0		0	0
0	0		0		
0	930.429	4636	0	0	0
0		0	0		
0	0		0		893.167

0	0	0	0
0		0	0
3035.5	871.167		503.083
0		0	0
0		558.538	329.154
0		369.611	684
403.25	0		0
0	0		0
0	0	0	0
Arena Green / SAP Center		0	0
0	1506.11	4942.16	0
0		0	0
0		0	0
0	0	0	0
0	0		27409
0	0		0
0	1101.43	1921.6	0
0		0	0
0		0	0
0	0	0	737.857
0	0	0	0
0		0	0
683.25	916	788.25	0
0		0	0
0		547	798.353
0		354.688	1171.33
247.688	0		0
0	0	0	745.8
0	0	0	0
0	0	0	0
Beale at Market		360.521	608.08
821.229	0	0	7148.71
639.268		0	0
618.767		518.85	319.143
0	224.174	447.825	331.593
595.736	391.78	0	0
662.953	448.297		3224.84
334.913	0	0	727.525
732.313	1302.71		0
0	0		0
2591.64	493.472	429.654	0
0		0	0
0	0	0	0
0		761.388	
785.916		843.8	0
0	0		0
0	0	0	0
970.458	522.783	0	0
1404.5	562.866		1054.47

0	446.977	446.381	
Broadway St at Battery St	712.513	914.102	761.27
1050.56	0	0	329.909
2841.56	0		0
1228.43	284.93	509.957	
0	228.985	571.551	655.532
937.498	1305.9	0	0
1476	1267.74	805.024	
548.812	0	2048.16	1125.86
492.553	467.599	0	0
0	0	0	0
604.157	1169.88	576.447	0
0	0	0	
0	0	0	0
0	961.47		
943.051	3093.12	0	
0	0	0	
0	0	0	0
1029.45	545.194	0	0
411.472	495.851		1196.57
0	2094.69	763.4	
California Ave Caltrain Station	0	0	0
0	0	0	0
0	11436.8	1645	
0	0	0	
5748.92	0	0	0
0	0	0	0
0	0	0	
0	0	0	0
0	0	1844	
2441.75	8172.94	615.516	0
0	0	0	0
0	0	1386	
0	0	0	3356.27
1384.14	0		0
0	0	0	0
0	0	0	0
0	0	0	0
2872	0	0	
0	1595.12	0	0
Castro Street and El Camino Real	0	0	0
0	0	0	0
0	0	6883.92	
0	0	0	
0	0	0	0
0	0	6830.44	0
0	0	0	

```
| 0          | 0          | 0          | 0          | 0          | 0
| 0
| 5499.53     | 0          | 0          | 475.892    | 2768       | 0
| 0          | 0          | 0          | 0          | 2544.7     | 0
| 0          | 0          | 0          | 0          | 1112.33    | 0
| 117967     | 0          | 0          | 0          | 0          | 0
| 0          | 0          | 0          | 0          | 0          | 0
| 0          | 0          | 0          | 0          | 0          | 0
| 0          | 0          | 0          | 0          | 0          | 0
| 0          | 0          | 0          | 0          | 0          | 0
| 0          | 4150       | 0          | 0          | 0          | 0
... (60 rows omitted)
```

1.3 3. San Francisco City Employee Salaries

This exercise is designed to give you practice with using the Table methods `.pivot` and `.group`. Here is a link to the [Python Reference](#) in case you need a quick refresher. The [Table Function Visualizer](#) may also be a helpful tool.

The data source we will use within this portion of the homework is [publicly provided](#) by the City of San Francisco. We have filtered it to retain just the relevant columns and restricted the data to the calendar year 2019. Run the following cell to load our data into a table called `full_sf`.

```
[37]: full_sf = Table.read_table("./DS/sf2019.csv")
      full_sf.show(5)
```

<IPython.core.display.HTML object>

The table has one row for each of the 44,525 San Francisco government employees in 2019.

The first four columns describe the employee's job. For example, the employee in the third row of the table had a job called "IS Business Analyst-Senior". We will call this the employee's *position* or *job title*. The job was in a Job Family called Information Systems (hence the IS in the job title), and was in the Adult Probation Department that is part of the Public Protection Organization Group of the government. You will mostly be working with the `Job` column.

The next three columns contain the dollar amounts paid to the employee in the calendar year 2019 for salary, overtime, and benefits. Note that an employee's salary does not include their overtime earnings.

The last column contains the total compensation paid to the employee. It is the sum of the previous three columns:

$$\text{Total Compensation} = \text{Salary} + \text{Overtime} + \text{Benefits}$$

For this homework, we will be using the following columns: 1. **Organization Group**: A group of departments. For example, the Public Protection Org. Group includes departments such as the Police, Fire, Adult Protection, District Attorney, etc. 2. **Department**: The primary organizational

unit used by the City and County of San Francisco. 3. **Job**: The specific position that a given worker fills. 4. **Total Compensation**: The sum of a worker's salary, overtime, and benefits in 2019.

Run the following cell to select the relevant columns and create a new table named `sf`.

```
[38]: sf = full_sf.select("Job", "Department", "Organization Group", "Total_
      ↪Compensation")
      sf.show(5)
```

<IPython.core.display.HTML object>

We want to use this table to generate arrays with the job titles of the members of each **Organization Group**.

Question 1. Set `job_titles` to a table with two columns. The first column should be called **Organization Group** and have the name of every “Organization Group” each listed only once in this column, and the second column should be called **Jobs** with each row in that second column containing an *array* of the names of all the job titles within that “Organization Group”. Don’t worry if there are multiple of the same job titles. **(9 Points)**

Hint 1: Think about how `group` works: it collects values into an array and then applies a function to that array. We have defined two functions below for you, and you will need to use one of them in your call to `group`.

Hint 2: You might need to rename one of the columns.

```
[46]: # Pick one of the two functions defined below in your call to group.
def first_item(array):
    '''Returns the first item'''
    return array.item(0)

def full_array(array):
    '''Returns the array that is passed through'''
    return array

# Make a call to group using one of the functions above when you define_
↪job_titles
job_titles = sf.group("Organization Group", collect=full_array).relabelled("Job_
↪full_array", "Jobs").select("Organization Group","Jobs")
job_titles
```

```
[46]: Organization Group | Jobs
Community Health | ['Painter Supervisor 1' 'Painter'
'Painter' ..., 'Nursin ...
Culture & Recreation | ['Electrician' 'Executive Secretary
2' 'Bldgs & Grounds ...
General Administration & Finance | ['Painter' 'Painter' 'Electrician'
..., 'Investigator, T ...
Human Welfare & Neighborhood Development | ['Dept Head I' 'Administrative
Analyst' 'Community Devel ...
```

```
Public Protection | ['IS Trainer-Journey' 'IS Engineer-
Assistant' 'IS Busine ...
Public Works, Transportation & Commerce | ['Heavy Equip Ops Asst Sprv' 'Heavy
Equipment Ops Sprv'
...
```

Question 2. At the moment, the `Job` column of the `sf` table is not sorted (no particular order). Would the arrays you generated in the `Jobs` column of the previous question be the same if we had sorted alphabetically instead before generating them? Explain your answer. To receive full credit, your answer should reference *how* the `.group` method works, and how sorting the `Jobs` column would affect this. (8 Points)

Note: Two arrays are the **same** if they contain the same number of elements and the elements located at corresponding indexes in the two arrays are identical. An example of arrays that are NOT the same: `array([1,2]) != array([2,1])`.

```
Table.group()
            .sf('Job')
            .group
```

Question 3. Set `department_ranges` to a table containing departments as the rows, and the organization groups as the columns. The values in the rows should correspond to a total compensation range, where range is defined as the **difference between the highest total compensation and the lowest total compensation in the department for that organization group**. (9 Points)

Hint: First you'll need to define a new function `compensation_range` which takes in an array of compensations and returns the range of compensations in that array.

```
[51]: # Define compensation_range first
def compensation_range(array):
    '''Returns the range of compensations in the array'''
    return max(array) - min(array)

department_ranges = sf.pivot("Department", "Organization Group", values="Total_
↪Compensation", collect=compensation_range)
department_ranges
```

```
[51]: Organization Group | Academy Of Sciences | Administrative
Services | Adult Probation | Airport Commission | Art Commission | Asian Art
Museum | Assessor | Board Of Appeals | Board Of Supervisors | Building
Inspection | Child Support Services | Children & Families Commission | Children
Youth & Families | City Attorney | City Planning | Civil Service Commission |
Controller | Department Of Public Works | Department of Technology | Dept Status
of Women | Dept of Emergency Management | Dept of Police Accountability |
District Attorney | Economic Workforce Development | Environment | Ethics
Commission | Fine Arts Museum | Fire Department | Health Service System |
Homeless Services | Human Resources | Human Rights Commission | Human Services |
Juvenile Court | Law Library | Mayor | Municipal Transportation Agcy | Police |
Port | Public Defender | Public Health | Public Library | Public Utilities
```


231712	0	0	290205	0
0	0	0	316098	0
0	0	0	0	0
0	301206	418740	0	0
0	0	0	0	0
0	0	0	0	0
0	293132	0	0	0
0	0	0	0	0
Public Protection	0	0	0	0
303419	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	325333	332041	0	0
335621	0	0	0	0
0	492128	0	0	0
0	0	0	342453	0
0	0	0	440552	0
0	0	0	0	337095
0	0	0	545324	0
320807	0	0	0	0
Public Works, Transportation & Commerce	0	0	0	0
0	445092	0	0	0
243582	0	340852	0	0
0	0	0	0	0
0	0	374263	0	0
0	0	0	0	0
0	300888	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	381639	0	393270	0
0	0	489773	0	0
0	0	0	0	0
0	0	0	0	0

```
[50]: # Define compensation_range first
def compensation_range(array):
    '''Returns the range of compensations in the array'''
    return max(array) - min(array)
```

Question 4. Why might some of the row values be 0 in the `department_ranges` table from the previous question. (8 Points)

department_ranges	0	Organization Group	Department	Total Compensation
compensation_range	(max(array) - min(array)).			0
(Organization Group Department)			(range) 0	

Question 5. Find the number of departments appearing in the `sf` table that have an average total

compensation of greater than 125,000 dollars; assign this value to the variable `num_over_125k`. (9 Points)

Note: The variable names provided are meant to help guide the intermediate steps and general thought process. Feel free to delete them if you'd prefer to start from scratch, but make sure your final answer is assigned to `num_over_125k`!

```
[56]: depts_and_comp = sf.select("Department", "Total Compensation")
      department_avg = depts_and_comp.group("Department", np.mean)
      num_over_125k = department_avg.where("Total Compensation mean", are.
      ↪above(125000)).num_rows
      num_over_125k
```

[56]: 23

1.4 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a pdf file for you to submit. **Please save before exporting!**

```
[59]: # should change the directory and file name matching to yours
      !jupyter nbconvert './DS/lab09(SSU).ipynb' --to pdf
```

```
[NbConvertApp] Converting notebook ./DS/lab09(SSU).ipynb to pdf
[NbConvertApp] Writing 131939 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | bibtex had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 118712 bytes to DS/lab09(SSU).pdf
```